

Bulk and Interfacial Behavior of Ionic Liquids from Molecular Dynamics Simulations

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Results of molecular dynamics simulations using many-body polarizable APPLE&P force field are presented for 34 ionic liquids covering a broad range of cations and anions. The relation between ionic liquid transport properties (ion self-diffusion coefficient and conductivity), thermodynamic properties (molar volume and heat of vaporization) and cation-anion binding energies are presented. Development of the nonpolarizable united atom force field for alkylpyrrolidinium and tetraalkyl ammonium-based ionic liquids is reported. The nonpolarizable force field yielded higher activation energy for ion transport as a function of temperature and slightly lower heat of vaporization compared to MD simulation results using polarizable APPLE&P force field. Interfacial properties of *N*-methyl-*N*-propylpyrrolidinium bis(trifluoromethane)sulfonyl imide ([pyr₁₃][TFSI]) near a graphite electrode and confined in nanopores have been studied as a function temperature, electrode potential or the size of the nanopore. A camel shaped differential capacitance as a function of applied potential is reported for the flat surface electrode and is correlated with the changes of the interfacial structure.

Introduction

Room temperature ionic liquids (ILs) have negligible vapor pressure, good thermal and electrochemical stability, good dissolution properties with many organic and inorganic compounds, low flammability(1). ILs are investigated as alternative materials for variety of applications from solvents for synthetic and catalytic applications,(2) to additives and electrolytes for lithium batteries,(3-6) actuators,(7, 8) reaction media and replacement of conventional solvents,(9) and active pharmaceutical ingredients(10). Importantly, IL properties can be tailored for specific chemical (separation, catalysis, reactions, propellants, explosives) or electrochemical (battery, actuators) applications by tuning the combination of cations and anions to achieve the desired thermodynamic, solvating and transport properties as well as safety or oxygen-nitrogen balance. However, screening a large number of possible cation/anion combinations also presents an enormous challenge for product design as synthesis and characterization of a large number of ILs is expensive. Efficient and reliable predictive tools can speed up the development cycle not only by providing expedient predictions of properties for specific ILs and IL mixtures, but also by providing an improved fundamental understanding of ILs and data needed for the development of semi-empirical structure-property relationship models.

In this manuscript we reported a study of the dependence of the ionic liquid transport properties (ion self-diffusion coefficient and conductivity) on the ionic liquid thermodynamic properties (molar volume and heat of vaporization) and cation-anion binding energies. Our previous report(11) on this topic has been extended to include dicyanotriazolate (DCTA⁻) and (CF₃)₂PO₂⁻ anions, semifluorinated alkane (F(CF₂)_n-CH₂-CH₂-) modified imidazolium cations. An improved APPLE&P force field (version fle17) for cyano containing anions N(CN)₂⁻=dca, C(CN)₃⁻ and B(CN)₄⁻ as been used. The revised sp-carbone parameters in the fle17 version of the force field improved description of the density of 2-butyne from 3% to 0.8% compared to experimental value.(12) An adequate heat of vaporization description for 2-butyne was observed (within 3 % of experiments) (12, 13). The nitrogen atom (in cyano groups) repulsion-dispersion parameters were adjusted to yield density and heat of vaporization of acetonitrile (CH₃CN). Substitution of these sp-carbon and cyano group nitrogen parameters to IL force field yielded density of [emim][N(CN)₂] of 1090 kg/m³ at 298 K for fle17 version of the force field compared the previously reported value of 1077 kg/m³ reported for the fle1 version of APPLE&P(14).

Relation Between Ion Transport, Heat Of Vaporization and Molecular Volume of Ionic Liquids

Molecular dynamics (MD) simulations followed a previously published protocol(14) and were performed for 34 ionic liquids consisting of cations and anion shown in Figure 1. A simulation box consisted of 125-216 ion pairs. APPLE&P force field from ref.(14) has been used with the modifications listed above. The Ewald summation method was used for electrostatic interactions between partial charges with partial charges and partial charges with induced dipole moments using k^3 from 6³ to 8³ k-vectors, and α from 8.5 - 9.9 Å. The interaction between an induced dipole and a partial charge separated by 3 bonds was scaled by 0.8, providing an improved description of the electrostatic potential around the molecules. Multiple timestep integration with an inner timestep of 0.5 fs (bonded interactions), a central time step of 1.5 fs for all nonbonded interactions within a truncation of 7.0 Å and an outer timestep of 3.0 fs for all nonbonded between 7.0 Å and the nonbonded truncation distance of 10.5 or 11 Å as well as for the reciprocal part of Ewald was employed. A Nose-Hoover thermostat (NPT and NVT simulations) and a barostat (NPT simulations) were used to control the temperature and pressure with the associated frequencies of 10⁻² and 0.5x10⁻³ fs. Induced dipoles were calculated via a direct iteration with a predictor method. Pressure tensor and virial were calculated using atom-based with bond (constrained) forces.

Verevkin(15) has successfully correlated H_{vap} with a term containing IL surface tension and molar volume for fifteen ILs. While many heat of vaporization values were reported from MD simulation studies,(16-26) and experiments(15, 27-32) most of them dealt only with a relatively small number of ILs and did not generate enough data to extensively consider relations between heat of vaporization (H_{vap}) and transport properties. We follow our previous report(11) and investigate the relationship between IL transport properties and H_{vap} , the cation-anion binding energy and ion sizes for an extended list of 34 simulated ionic liquids in listed in Table I.

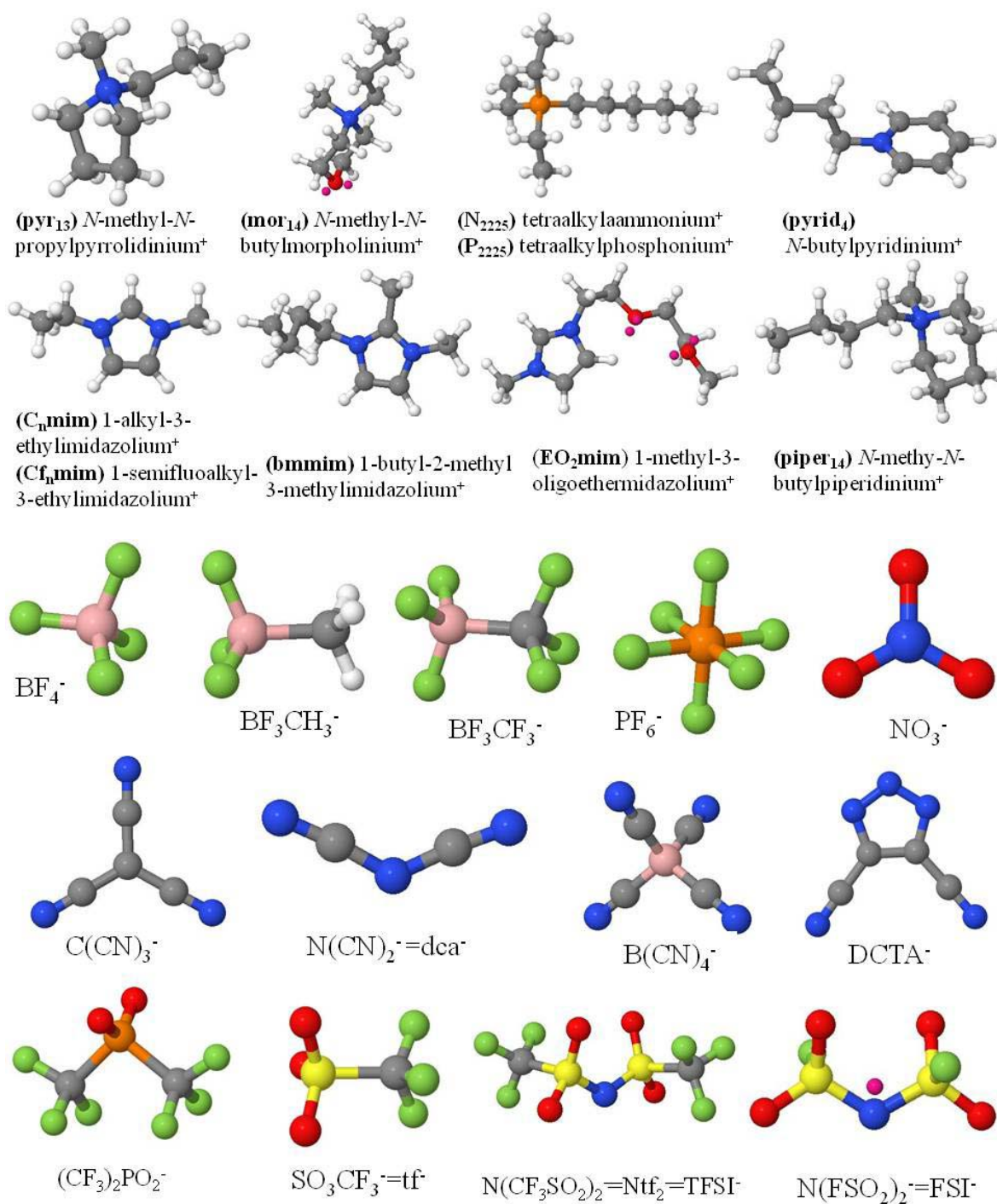


Figure 1. A list of anions and cations used in MD simulations.

TABLE I. A list of simulated ionic liquids (see Figure 1 for ion notation).

IL #	cation	anion	IL #	Cation	Anion
1	emim	CF ₃ BF ₃	18	piperid ₁₄	TFSI
2	emim	BF ₄	19	morph ₁₄	TFSI
3	bmim	BF ₄	20	emim	SO ₃ C ₄ H ₉
4	bmim	PF ₆	21	emim	FSI
5	pyrid ₄	BF ₄	22	pyr ₁₃	FSI
6	emim	CH ₃ BF ₃	23	emim	N(CN) ₂
7	emim	TFSI	24	bmim	N(CN) ₂
8	bmim	TFSI	25	pyr ₁₂	N(CN) ₂
9	EO ₂ mim	TFSI	26	emim	C(CN) ₃
10	bmmim	TFSI	27	phosph ₁₂	C(CN) ₃
11	pyr _{1,102}	TFSI	28	emim	B(CN) ₄
12	pyrid ₄	TFSI	29	bmim	DCTA
13	c6mim	TFSI	30	bmim	NO ₃
14	pyr ₁₃	TFSI	31	Cf ₂ Ch ₂ -mim ^a	TFSI
15	pyr ₁₄	TFSI	32	Cf ₄ Ch ₂ -mim ^a	TFSI
16	bmim	CF ₃ SO ₃	33	Cf ₆ Ch ₂ -mim ^a	TFSI
17	N ₁₁₁₄	TFSI	34	bmim	(CF ₃) ₂ PO ₂

^a Cf- denotes CF₃- or CF₂- group, while Ch – denotes CH₂ group.

Ionic conductivity was calculated following the previously discussed(14) procedure from ion self-diffusion coefficients and degree of dynamic ion correlation. The relation between ionic conductivity and H_{vap} is shown in Figure 2. A significant correlation has been observed in Figure 2. A similar quality correlation was observed between the average diffusion coefficients of ions D and H_{vap} (not shown). One might argue that not only heat of vaporization but the cation-anion binding energy ($E_{+/-}$) could be correlated with ion transport in ILs. While no correlation between the cation-anion binding energy ($E_{+/-}$) and ion diffusion coefficients was found, an improved correlation was observed between $-\log(D)$ and a combination of H_{vap} and $E_{+/-}$ by adding a fraction of the cation-anion binding energy as shown in Figure 3. The sum $H_{\text{vap}} + 0.18 E_{+/-}$ could be thought of as an effective cohesive energy that reflects contributions due to removing an ion pair from IL and breaking it apart. The largest deviation in Figure 3 is observed for [bmim][(CF₃)₂PO₂] followed by two ILs containing bmmim⁺ and EO₂mim⁺ cations (ILs #9 and #10). While the reason for the large deviation of [bmim][(CF₃)₂PO₂] is currently under investigation, methylation of C₍₂₎ hydrogen (bmim⁺ → bmmim⁺) significantly changes the location of the Ntf₂⁻ anion near C₍₂₎ carbon of bmim⁺ cation, thus changing the pathway for relaxation associated with the anion moving over C₍₂₎-H group. This changes the diffusion mechanism and results in a slower diffusion in [bmmim][Ntf₂] compared to [bmim][Ntf₂] despite slightly (by 0.7 kcal/mol) lower H_{vap} for [bmmim][Ntf₂] compared to [bmim][Ntf₂].

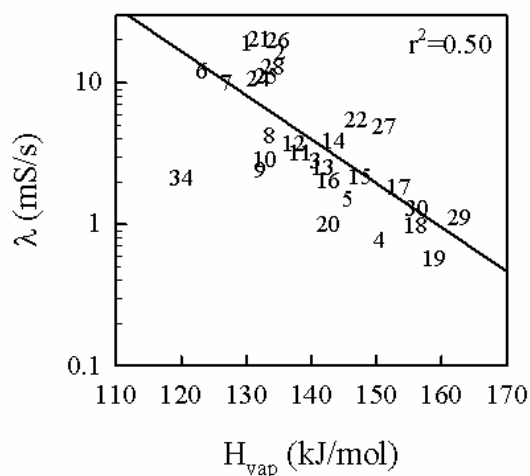


Figure 2. A relation between ionic conductivity and heat of vaporization for ionic liquids listed in Table I.

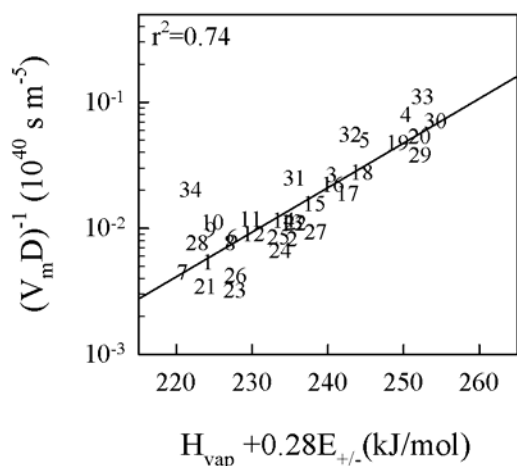


Figure 3. A relation between ionic conductivity, molar volume (V_m), heat of vaporization and cation-anion binding energy ($E_{+/-}$) for ionic liquids listed in Table I.

Development of United Atom Nonpolarizable Force Field for Ionic Liquids

While MD simulations using many-body polarizable APPLE&P force field(14, 33, 34) accurately predicted density, heat of vaporization, ion self-diffusion coefficient, conductivity and viscosity and crystal lattice parameters of more than 30 ILs, it is highly desirable to develop less computationally expensive nonpolarizable and united atom force fields. We recently utilized force matching (FM) methodology to understand the influence of polarization on the water, organic liquids, ionic liquid structure and transport and developed numeric nonpolarizable all atom force fields.(35) We found that for ionic liquids, unlike water, where the FM approach was found to yield an accurate representation of the liquid phase structure predicted by a polarizable model, the FM approach does not result in a two-body potential that accurately reproduces either

structure or dynamics predicted by the polarizable IL model. We also found that two-body force field yielded a slower dynamics in ionic liquids. In this contribution we take an empirical approach to development of even less computationally expensive united atom nonpolarizable force field that later will be used in studies of electrochemical interfacial behavior of ionic liquids.

A nonpolarizable united atom (UA) force field for *N*-methyl-*N*-propylpyrrolidinium (pyr_{13}^+) bis(trifluoromethane) sulfonamide (TFSI) has been developed using all atom (AA) APPLE&P (Atomistic Polarizable Potential for Liquids, Electrolytes and Polymers)(14, 34) as a starting point. In the process of UA force field development for $[\text{pyr}_{13}][\text{TFSI}]$ we also developed a nonpolarizable UA force fields for n-alkanes (validating it for C_5H_{12} , $\text{C}_{10}\text{H}_{22}$) and two ionic liquids $[\text{N-methyl-}N\text{-butylpyrrolidinium } (\text{pyr}_{14}^+)][\text{TFSI}]$ and $[(\text{CH}_3)_3\text{N}^+(\text{C}_4\text{H}_9)][\text{TFSI}]$ denoted as $[\text{N}_{4111}][\text{TFSI}]$. These additional compounds (n-alkanes, $[\text{N}_{4111}][\text{TFSI}]$) were needed to derive the non-bonded repulsion-dispersion parameters for $[\text{pyr}_{13}][\text{TFSI}]$, while $[\text{pyr}_{14}][\text{TFSI}]$ was used to perform an additional force field validation. Specifically, the terminal methyl group and ethylene carbon repulsion-dispersion parameters were derived from n-alkane density and heat of vaporization, while the repulsion-dispersion parameters for the terminal methyl group attached to the N^+ in ionic liquid (IL) cations were largely based on the density, heat of vaporization and ion self-diffusion coefficients of $[\text{N}_{4111}][\text{TFSI}]$ as discussed below.

The force field parameterization proceeded in the following way. First, partial charges for TFSI⁻, N_{4111}^+ and pyr_{13}^+ were fit to describe the electrostatic potential on a grid of points around molecules, as well as molecular gas phase dipole moment, all obtained from QC calculations at MP2/aug-cc-pvDz level for cations and MP2/cc-pvTz for TFSI. Second, bond lengths and natural bending angles were taken from APPLE&P without modification. Third, dihedral angle parameters were determined by fitting the gas phase conformational-energy surface of model molecules as determined from quantum chemistry as previously described for APPLE&P.(14) Fourth, the repulsion-dispersion parameters (see Table II) were developed. The $\text{C}_U \dots \text{C}_U$ and $\text{C}_{mU} \dots \text{C}_{mU}$ repulsion-dispersion parameters were fit to density and heat of vaporization of C_5H_{12} and $\text{C}_{10}\text{H}_{22}$ as shown in Table II. Next, $\text{C}_N \dots \text{C}_N$ and $\text{C}_{mN} \dots \text{C}_{mN}$ repulsion-dispersion parameters were fit to $[\text{pyr}_{13}][\text{TFSI}]$ and $[\text{N}_{4111}][\text{TFSI}]$ density and self-diffusion coefficient from experiments and heats of vaporization obtained from MD simulations using APPLE&P force field as shown in Table III. The TFSI S...S repulsion-dispersion interactions were adjusted to improve ion self-diffusion coefficients for $[\text{pyr}_{13}][\text{TFSI}]$, $[\text{pyr}_{14}][\text{TFSI}]$ and $[\text{N}_{4111}][\text{TFSI}]$ as a function of temperature. We could not simultaneously obtain a good fit for ion self-diffusion coefficient and heat of vaporization, therefore, we settled on the compromise when MD simulations using UA force field yielded ion-self diffusion coefficients lower compared to experiments and MD simulations using APPLE&P, while heat of vaporization was slightly below the results of MD simulations with APPLE&P.

TABLE II. The repulsion-dispersion (R/D) types used in the nonpolarizable force field.

Label	atom R/D types
C _U	sp ³ carbon in - CH ₂ - group
C _{mU}	sp ³ carbon at chain ends (CH ₃ -, CF ₃ - groups) not connected to N ⁺
C _m	sp ³ carbon at chain ends (CF ₃ - groups), fluorine atoms were not callapsed
C _N	sp ³ carbon in - CH ₂ - group connected to N ⁺
C _{Nm}	sp ³ carbon at chain ends (CH ₃ -, CF ₃ - groups) connected to N ⁺
N ⁺	nitrogen with the formal charge +1e
O=	all oxygen atoms (O=S)
F	all fluorine atoms
S ⁻	sulfur in TFSI ⁻

TABLE III. Thermodynamic properties (density, heat of vaporization, self-diffusion coefficient) of n-alkanes from MD simulations (ρ^{MD} , $H_{\text{vap}}^{\text{MD}}$, D^{MD}) using united atom force field, experiments (ρ^{exp} , $H_{\text{vap}}^{\text{exp}}$, D^{exp}) and deviation of MD results from experiments ($\Delta\rho$, ΔH_{vap} , ΔD) in per cents.

Liquid	ρ^{MD} (ρ^{exp}) $\Delta\rho(\%)$ (kg m ⁻³)	$H_{\text{vap}}^{\text{MD}}$ ($H_{\text{vap}}^{\text{exp}}$) $\Delta H_{\text{vap}}(\%)$ (kJ/mol)	D^{MD} (D^{exp}) $\Delta D(\%)$ (10 ⁻¹⁰ m ² /s)	references
C ₅ H ₁₂	628 (621) 1.1	26.7 (26.7) -0.3	65.5 (56.2) 17	(36),(36),(37)
C ₁₀ H ₂₂	726.7 (724.7) 0.3	51.9 (51.4) 0.9	22.7 (14) 62	(36), (36),(38)

We begin comparison of the nonpolarizable UA and polarizable AA (APPLE&P) force fields by analyzing n-alkane data shown in Table III. We observe that for the same density and heat of vaporization UA model predicts faster by 16% and 67% self-diffusion coefficients for C₅H₁₂ and C₁₀H₂₂, respectively. Note that MD simulations using UA models using Lennard-Jones 12-6 repulsion-dispersion form predicted C₁₀H₂₂ self-diffusion coefficient only 10% faster than experiments(38). The slower diffusion for the Lennard-Jones 12-6 UA model compared to the currently used exp-6 is attributed to the steeper repulsive part of LJ 12-6 form compared to exp-6. We could have increased λ steepness parameter beyond values of 14-16, typically used in APPLE&P, thus, decreasing self-diffusion coefficients for n-alkanes and brining them close to experiments. We, however, have not done this because at high λ ($\lambda > 20$) ion diffusion in ILs would significantly decreased and description of IL transport would have deteriorated.

Comparison of density, heat of vaporization and ion self-diffusion coefficients from MD simulations using UA model and APPLE&P is shown in Table IV. The UA model was successfully fit to match experimental and APPLE&P densities. Description of the ion self-diffusion coefficient using UA force field as a function of temperature was less successful, however. At high temperature (393 K) agreement between MD simulations using UA force field and experiments is generally good, with MD simulations predicting self-diffusion coefficients only 8-38% lower. In the simulations reported here no finite

size correction to ion self-diffusion coefficients was applied because IL viscosity needed for calculating this correction was not calculated.(14) Magnitude of the finite size correction is expected to be around 10% for the simulated ILs and box sizes. Application of this correction will reduce the disagreement between MD simulations using UA force field and extrapolated experimental data to below 28% at 393 K. As temperature decreases to 298 K description of the ion self-diffusion coefficient from MD simulations using UA force field deteriorates and we observed a disagreement of a factor of 2-4 with experiments and with results using APPLE&P force field. Note that MD simulations using APPLE&P force field yielded ion transport in excellent agreement with experiments.(14) Higher activation energy for the diffusion coefficient temperature dependence of the UA nonpolarizable force field vs. polarizable AA (APPLE&P) force field is attributed to the absence of polarization in the UA force field as previously discussed.(14, 35) Despite this deficiency of UA nonpolarizable force field in describing temperature dependence of ion dynamics, we feel that the developed force field is a significant improvement over the previously published work for alkylpyrrolidinium and alkylammonium-based ILs. Indeed, Tsuzuki et al.(39) predicted ion diffusion of $[N_{4111}][TFSI]$ and $[pyr_{14}][TFSI]$ a factor of 20-40 slower than experiments at 353 K. Siqueira et al.(40) predicted ion diffusion of $[N\text{-ethyl-}N,N\text{-dimethyl-}N\text{-(2-methoxyethyl)ammonium}][TFSI]$ about 1 order of magnitude lower than experiments.

TABLE IV. Thermodynamic properties of ionic liquids from MD simulations (ρ^{MD} , H_{vap}^{UA} , D^{MD}) using united atom force field, all atom force field H_{vap}^{AA} , experiments (ρ^{exp} , D^{exp}).

	T (K)	ρ^{MD} (ρ^{exp}) (kg m ⁻³)	H_{vap}^{UA} (H_{vap}^{AA}) (kJ/mol)	D_+^{MD} (D_+^{exp}) (10 ⁻¹⁰ m ² /s)	D_-^{MD} (D_-^{exp}) (10 ⁻¹⁰ m ² /s)	references
$[N_{4111}][TFSI]$	393	1325		1.36 (2.22)	1.36 (2.07)	
$[N_{4111}][TFSI]$	333	1373 (1357)		0.20 (0.538)	0.21 (0.487)	(41)
$[N_{4111}][TFSI]$	298	1406 (1393)	145.9 (157.1)	0.031 (0.131)	0.035 (0.117)	(41)
$[pyr_{13}][TFSI]$	393	1341 (1343)		2.2	2.0	(41-43)
$[pyr_{13}][TFSI]$	333	394 (1384)		0.43	0.42	(42, 43)
$[pyr_{13}][TFSI]$	298	1426 (1408-1447)	140.0 (143.6)	0.086 (0.26)	0.085 (0.17)	(42)
$[pyr_{14}][TFSI]$	393	1297		1.6 (2.3)	1.9 (2.1)	(41)
$[pyr_{14}][TFSI]$	333	1353 (1367)		0.29 (0.64)	0.33 (0.53)	(41)
$[pyr_{14}][TFSI]$	298	1384 (1399)	143.2 (147.6)	0.038 (0.177)	0.051 (0.142)	(41)

Ability of the developed UA force field to describe ionic liquid structure was investigated by comparing the three dimensional (3-D) distribution function of the TFSI⁻ oxygen atom (denoted O(TFSI)) around pyr_{13}^+ and N_{4111}^+ cation shown in Figure 4 and 5, respectively. We observe that the isosurfaces of O(TFSI) around pyr_{13}^+ are slightly

shifted and occupy a significantly larger space. This behavior is consistent with the maximum of the 3-D distribution first peak being 56 for UA model and only 35 for APPLE&P. A significantly higher first maximum of 3-D distribution and larger occupied space for the isosurface are indicative of a stronger ordering of pyr_{13}^+ and TFSI^- in a liquid phase for UA model compared to AA model resulting in a slower dynamics for the former model. Somewhat similar picture was observed for $[\text{N}_{4111}][\text{TFSI}]$, with the maximum of 3-D distribution first peak being 47 for UA and 15 for AA APPLE&P, however, a larger discrepancy of 3-D isosurfaces was observed for $[\text{N}_{4111}][\text{TFSI}]$ compared to $[\text{pyr}_{13}][\text{TFSI}]$. Finally, heat of vaporization of the UA model was found to be somewhat lower than for APPLE&P as shown in Table IV. We conclude that the developed efficient nonpolarizable united atom force field adequately predicts ionic liquid density, heat of vaporization, structure and dynamics.

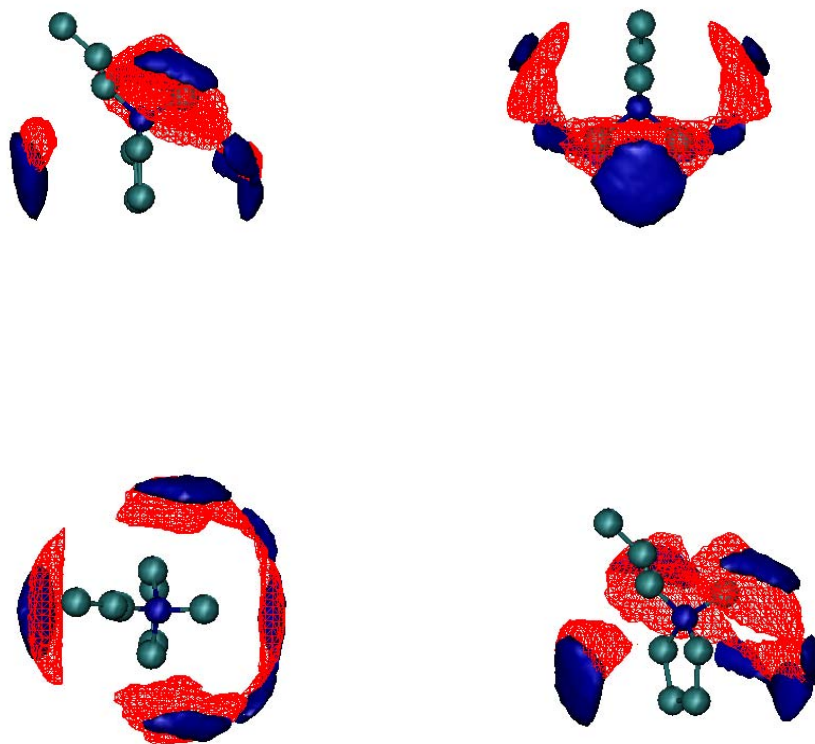


Figure 4. Isosurface of oxygen of TFSI^- around pyr_{13}^+ cation for $\rho/\rho^{\text{random}}=7$ (ρ -local number density), where red wireframe isosurface for UA force field, blue solid isosurface for APPLE&P force field.

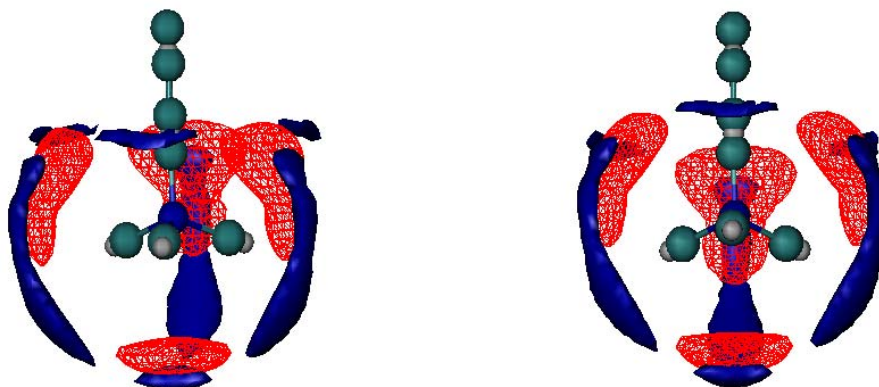


Figure 5. Isosurface of oxygen of TFSI⁻ around N4111⁺ cation for $\rho/\rho^{\text{random}}=8$ (ρ -local number density), where red wireframe isosurface for UA force field, blue solid isosurface for APPLE&P force field.

Structure and Capacitance of Ionic Liquid Multilayer in Capacitors

Experiments on electrolytes revealed complex camel shaped differential capacitance (DC) for both ILs and aqueous solutions of salts, consisting of a minimum near PZC, two maxima and decreasing wings of DC at larger positive or negative electrode potentials. Simulations using simple charged-sphere models(44, 45) predicted a maximum in DC at the PZC (leading to the so-called bell-shaped behavior of DC).(46, 47) The maximum is promoted by high density and high thermal energy (given as the ratio of kT and the electrostatic energy for ions in close contact). In contrast, low density and low thermal energy promote a minimum in DC near the PZC, leading to the so-called U-shaped DC. A complex dependence of the DC on electrode potential is observed in simple spherical models, indicating that complex molecular structure is not needed to produce camel-shaped DC, nevertheless, a *specific* behavior of a given ionic liquid/electrode combination depends upon the details of the molecular structure of the ions,(48) the ion-substrate interactions, and temperature. In order to understand the realistic (not a simplified charge sphere model) ionic liquid structure at the interface and capacitance as a function of applied potential we simulated [pyr₁₃][TFSI] sandwiched between graphite sheets as shown in Figure 6. The united atom nonpolarizable force field for [pyr₁₃][TFSI] developed above was used in simulations.

The simulations were performed under constant and controlled potential difference between electrodes(49, 50). For the purpose of this work we have chosen temperatures of 453 K, 393 K and 363 K, which are sufficiently high to allow structural equilibration in several nanoseconds. Thus, 24 ns production run simulations for each individual trajectory resulted in a reliable prediction of the interfacial structure and DC. The electrolyte density in the 20 Å slab in the middle of the simulation cell was set to be the same as of a 3D-periodic bulk [pyr₁₃][TFSI] the same temperature and pressure of 1 atm. The statistics on EDL structure were collected at any 30 fs over two sets of 12 ns simulations. The second set of 12 ns simulations was run to test reproducibility of our

results. We found that both trajectories predicted similar results. For each temperature, 58 trajectories were run at potential difference between electrodes in a range 0 V - 6.6 V.

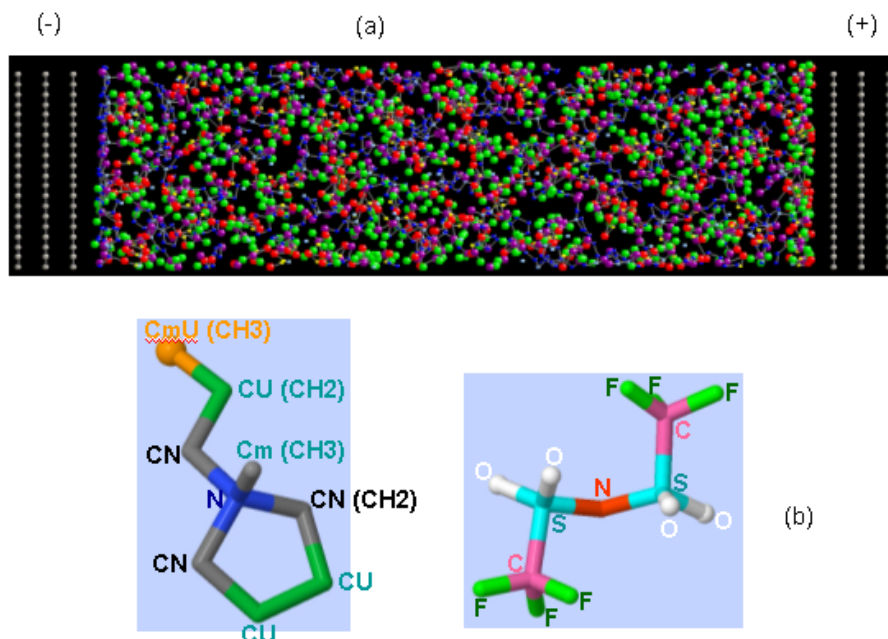


Figure 6. Image snapshot of a typical simulation setup at 4.2 V at 393 K for a 2D-geometry system. The electrode atoms are represented as grey spheres. The (-) and (+) sign indicate the polarity of the electrodes. The scale at the bottom of the figure indicates the distances along the direction normal to electrode surface (our OZ axes).

First, we calculated the potential of zero charge (PZC). PZC was approximated from the Poisson potential drop within the EDL at 0 V potential between the two electrodes. Values of PZC of -0.37 V, -0.34 V and -0.41 V for temperatures 363 K, 393 K and 453 K, respectively, were obtained. Negative PZC indicate a larger affinity of the uncharged electrode surface for the anions. The relatively large negative PZC is due to the high density of negatively charged fluorine atoms of TFSI situated next to the uncharged graphite electrode as shown in Figure 7 for 0 V. The potential drop within the electric double layer, V_{EDL} , is defined as the difference between the electrode potential and the screened bulk potential, $V_{EDL} = V_{\text{electrode}} - V_{\text{bulk}}$. Since the potential of zero charge (PZC) is non-zero for our systems, it is convenient to set the potential scale V_{RPZC} as relative to potential of zero charge, $V_{RPZC} = V_{EDL} - PZC = V_{\text{electrode}} - V_{\text{bulk}} - PZC$.

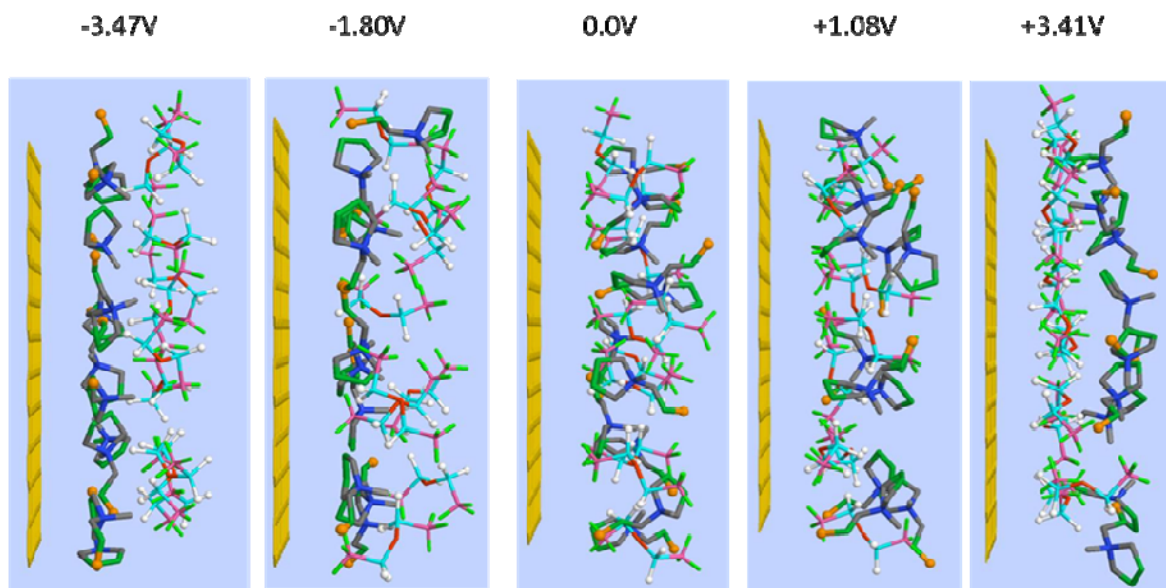


Figure 7. Structure of the double layer of $[\text{pyr}_{13}][\text{TFSI}]$ next to graphite at various potentials relative to PZC.

The differential capacitance was calculated as a function of applied voltage and is shown in Figure 8. At 363 K and 393 K the DC is camel-shaped, having a minimum shifted about -1 V from PZC and exhibiting saturation (maxima) at around $+1.5$ V_{RPZC} and -2.5 V_{RPZC} . A steep decrease in DC is observed for large positive and negative electrode potentials. The maximum in DC on the positive electrode is greater than that at the negative one). It is interesting to note that the PZC in our simulations corresponds to neither a minimum nor maximum in DC. While simple EDL theories predict a minimum in DC at PZC, experimental measurements of DC often show that the position of the minimum or maximum in DC depends upon the particular system and does not correspond to PZC.

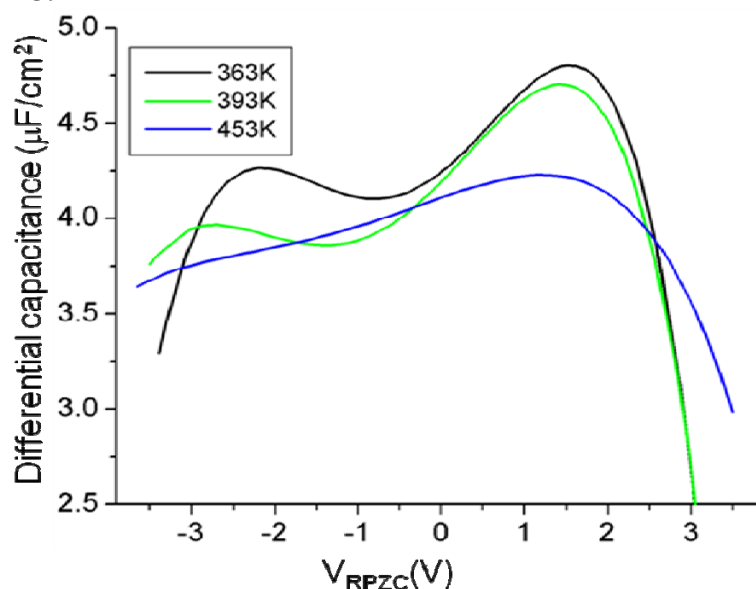


Figure 8. The differential capacitance as a function of the electrode potential relative to the potential of zero charge.

Next we discuss the relationship between $[\text{pyr}_{13}][\text{TFSI}]$ interfacial structure and differential capacitance shown in Figures 7 and 8. At the PZC, the inner electrolyte layer

is occupied largely by the propyl tail of pyr₁₃ (essentially non-polar) and the fluorine atoms of the TFSI (small negative charge) with a significant population of C carbons of pyr₁₃. These groups with the relatively small charge could be seen as “latent voids”. There are relatively few highly-charged TFSI oxygen and nitrogen atoms or pyr₁₃ N or C_N atoms in the inner layer. As V_{RPZC} becomes increasingly negative and the number of TFSI anions with their center-of-mass in the inner layer disappears and we observe an increase in TFSI in the intermediate layer until around $-1 V_{RPZC}$. In this range (0 to $-1 V_{RPZC}$), the electrode is becoming increasingly crowded as the density of positively charged pyr₁₃ rings increases and the fluorine atom density remains roughly constant in this range of potentials. At larger negative potentials, TFSI ions are pushed away from the surface and the pyr₁₃ ring rotating from being perpendicular to the surface at PZC to parallel to the surface. This results in a decrease in the footprint of the pyr₁₃ molecule allowing for a relatively high packing density before the onset of saturation effects. These occur at potentials (and associated electrode charge densities) more negative than about $-2.5 V_{RPZC}$. The observed reorientation of pyr₁₃ at the interface with increasing V_{RPZC} seem to correlate with the experimental evidence for 1-ethyl-3-methylimidazolium (emim⁺) cation orientation deduced from FT-IR and situ surface-enhanced infrared absorption spectroscopy(51-53). In line with our observations, Nanbu et al.(51-53) concluded that “emim⁺ cation present in the interphase orients vertically or, at least not parallel, with its methyl or ethyl group of emim⁺ directed toward the surface” in a potential range of -1.3 to $+0.6$ V vs. Ag|Ag⁺ and parallel to the surface at large negative potentials.

As V_{RPZC} becomes increasingly positive from 0 V_{RPZC} , Figure 7 reveals a dramatic increase in the population of highly negatively charged TFSI oxygen atoms next to graphite surface. With increasing V_{RPZC} the population of pyr₁₃ propyl end groups remains reasonably high even at the highest positive potentials when most pyr₁₃ center of mass moved out from the inner most layer to the intermediate layer and outer layer. The faster saturation of differential capacitance observed on the positive electrode (around $1.5 V_{RPZC}$) compared to the negative electrode (around $-2.5 V_{RPZC}$) is likely the result of the relatively large footprint of TFSI compared to pyr₁₃ and the propensity of the neutral pyr₁₃ propyl groups and, to a less degree, pyr₁₃ C atoms, to remain adsorbed on the interface at large positive potentials. In the potential range of 0 to $+1.5 V_{RPZC}$, the differential capacitance increases rapidly with increasing electrode potential, indicating that adsorption of TFSI oxygen atoms on the surface is occurring easily and brings highly charge oxygen atoms close to the surface forming a highly compact and charged inner layer open reorientation from perpendicular to parallel to the surface. However, at $+1.5 V_{RPZC}$, most TFSI in the interface they have all reoriented, and further increase of negative charge requires bringing new TFSI anions to the already crowded interface. Part of this crowding may come from the remaining propyl tails of the pyr₁₃ cations whose center-of-mass occupies the intermediate layer even at high positive potentials.

Structure of Nanoconfined Ionic Liquids

Next, we performed MD simulations of [pyr₁₃][TFSI] IL reservoir in contact with the nanoporous electrodes at 6.6 V. The simulation box setup is shown in Figure 9. The system consisted in two arrays of nanoporous electrodes, with the main axes of the tubes parallel to our OZ direction. The tubes were arranged in a hexagonal packed array along

xy-plane. The electrostatic potential of the nanotube C-atoms was controlled to a desired value by minimizing the total electrostatic energy of the system with respect to the electrode charges. The distance between nanotubes was set such that the ion insertion can be achieved only inside the tubes and not in the space between the tubes. The systems were prepared such that the density in the bulk electrolyte is the same as for a 3D-system at zero pressure. Nanotubes were arranged in such a way that no ion can penetrate between them as shown in Figure 9.

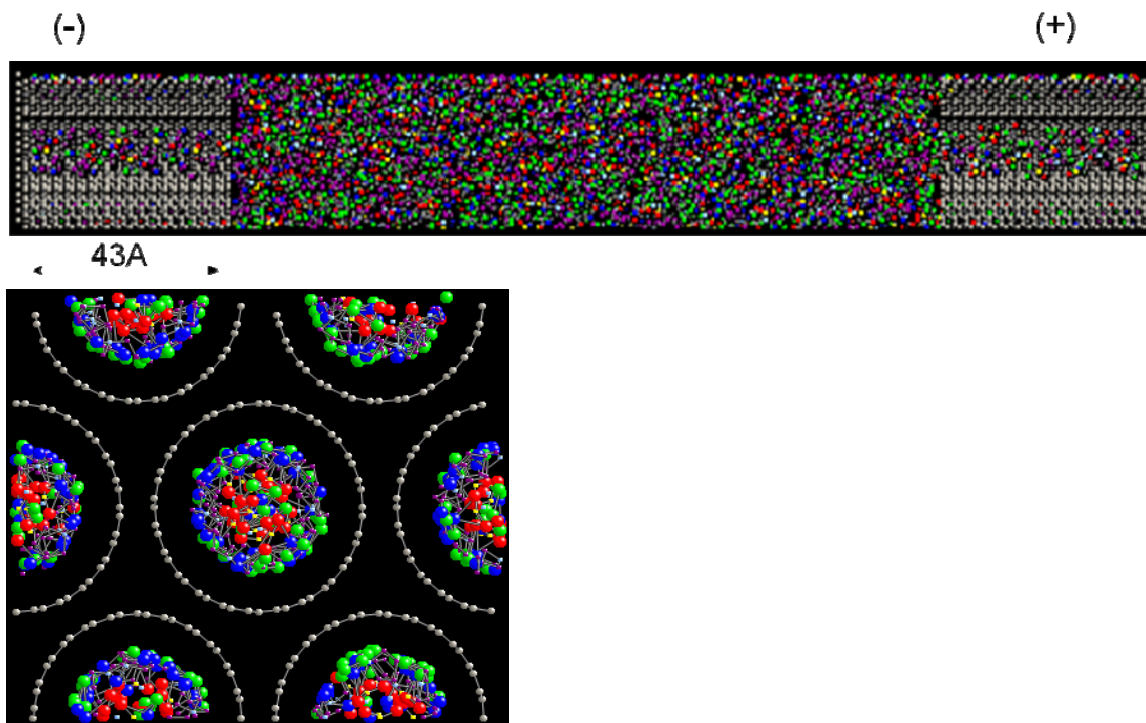


Figure 9. A snapshot of the simulation box.

Our preliminary results indicate that 9.49 Å and 10.84 Å only counterions fill nanopores with co-ions being excluded, while both counter and co-ions fill pores larger than 16 Å as shown in Figure 10. A well ordered ion arrangements were observed in all nanopores. In the larger nanopores a wrapped layers ionic liquid structure has been developed.

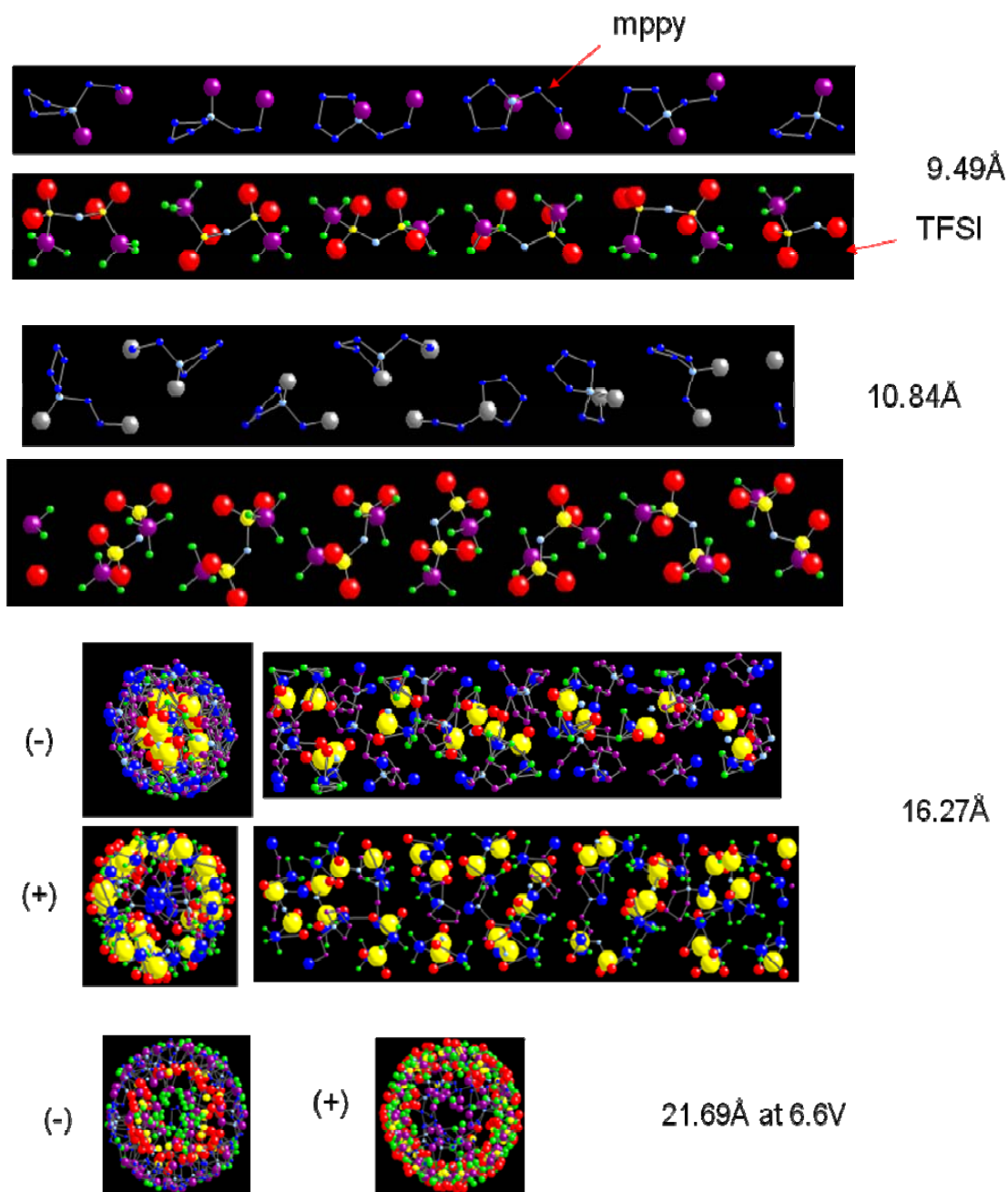


Figure 10. Structure of $[\text{pyr}_{13}][\text{TFSI}]$ confined in nanopores at 6.6 V total applied potential.

Conclusions

MD simulations of 34 ILs revealed a correlation between the effective cohesive energy of IL expressed as heat of vaporization and a fraction of the cation-anion binding energy and ion diffusion coefficients multiplied by volume. A camel shaped differential capacitance of $[\text{pyr}_{13}][\text{TFSI}]$ vs. applied voltage found and explained using changes of the interfacial layer structure. In simulations of the nanoconfined $[\text{pyr}_{13}][\text{TFSI}]$ at 6.6 V a

well ordered ion arrangements were observed. For the smallest pore sizes only counterions fill nanopores with co-ions being excluded, while both counter and co-ions fill pores larger than 16 Å.

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